

SliderMC Command Cheat Sheet

Firmware V1.0. Same groups as the [printable sheet](#). Canonical prose: [protocol.md](#). Dual-axis timing: [dual-movement.md](#).

Regenerate: python tools/render_command_cheatsheet.py

Call — send this (brackets = optional; bare = omit args).
Reply — typical success line; – = silent (errors still !E:...).

S — Set (session)				
SHORT	LONG	CALL	REPLY	DESCRIPTION
SS	SetSpeed	SS [<v>]	–	Cruise speed mm/s (≤ max_speed); bare reloads init_speed; live on next fill. Dual MT: axis1=session, axis2×ratio.
SA	SetAccel	SA [<a>]	–	Peak accel mm/s ² (≤ max_accel); bare reloads init_accel; live on next fill. Dual MT: same ratio scaling as SS.
SE	SetEnable	SE [0 1]	–	Driver enable 0 1; bare toggles; required before motion; off = hard stop.
ST	SetTerminal	ST [0 1]	–	Terminal Mode 0 1; bare toggles; local echo + UART sniff to USB (expert).
SV	SetVerbose	SV [0 1]	–	Verbose #... push 0 1; bare toggles; ~3 Hz (rate via verbose_rate_hz).
SD	SetDebug	SD [0..5]	–	USB-only debug level 0..5; bare restores default; never on UIC UART.

G — Get (session)				
SHORT	LONG	CALL	REPLY	DESCRIPTION
GS	GetSpeed	GS	GS:<mm/s>	Current session cruise speed.
GA	GetAccel	GA	GA:<mm/s ² >	Current session acceleration.
GE	GetEnable	GE	GE:0 1	Driver enable state.
GT	GetTerminal	GT	GT:0 1	Terminal Mode state.
GV	GetVerbose	GV	GV:0 1	Verbose push state.
GD	GetDebug	GD	GD:<0..5>	USB debug level.

I — Is / status				
SHORT	LONG	CALL	REPLY	DESCRIPTION
IM	IsMoving	IM	IM:0 1	Moving or settling on any active axis.
IH	IsHoming	IH	IH:0 1	Homing cycle active.
IL	IsLimit	IL	IL:0 1	At soft-limit position (axis1).
IE	IsError	IE	IE:0 1	PIN_DRV_ERROR / EMO latched.
IP	IsPosition	IP	IP:<pos> [<pos2>]	Axis-1 position; second field when axis2_use=1.
IA	IsAxis	IA	IA:1 2	Active axis count (config_axis2_enabled).
IT	IsTarget	IT	IT:<pos> –	Axis-1 seek target, or – if none / soft-stop.
IR	IsReady	IR	IR:0 1	1 only if idle, not homing, enabled, and not waiting.
IW	IsWaiting	IW	IW:0 1	1 if any W / WM / WH wait is active.
ID	IsDiag	ID	ID:underrun=N peak_hz=... overshoot=... fifo_min=...	Motion diag counters (FIFO underrun, peak STEP Hz, ...).
IZ	IsReset	IZ	IZ:<reason>	Last chip reset: power wdt run soft debug brownout ...
IX	Pinout	IX	(multi-line table)	ASCII GP / name / desc (≤80 cols). Axis-2 rows only if axis2 on.

M — Movement				
SHORT	LONG	CALL	REPLY	DESCRIPTION
MT	MoveTo	MT <pos> [<pos2>]	–	Absolute user units; optional 2nd axis; skip none/N/_/*; needs SE; live-retarget. Dual: time-sync ratio.
M	Move	M <delta> [<delta2>]	–	Relative move (alias MoveBy); same dual/skip rules as MT.
ML	MoveLeft	ML [0 1 2]	–	Jog –; mask 0=both, 1=axis1, 2=axis2 when axis2 on; soft-stop MS/!.
MR	MoveRight	MR [0 1 2]	–	Jog +; mask same as ML; soft-stop MS/!.
MH	MoveHome	MH [1 2]	–	Homing; axis 1 (default) or 2; no-op if home_mode=0; cancel MS/H.
MS	MoveStop	MS	–	Soft decelerate both axes; keeps enable; does not cancel waits. Dual: scaled accel kept.

P — Path				
SHORT	LONG	CALL	REPLY	DESCRIPTION
PC	PathClear	PC	–	Clear path buffer (count→0); !E:busy while PG active.
PD	PathData	PD <um> [<um2>]	–	Append signed μm sample(s); optional axis2; skip→0; OK while PG (live stream).
PG	PathGo	PG	–	Play buffer from sample 0; needs SE; !E:empty busy disabled. MS/H ends path.
PN	PathNumber	PN	PN: <count>	Samples in buffer; allowed during PG.
PS	PathSlice	PS [<us>]	–	Slice length μs (≥1000); bare→init_path_slice_us; !E:busy while PG.

X — Extender				
SHORT	LONG	CALL	REPLY	DESCRIPTION
X0-3	Ext0-3	Xn [0 1]	–	Ext out n logical 0 1; bare toggles; glued X00≡X0 0; OK during EMO. X4+ rejected.

C — Config				
SHORT	LONG	CALL	REPLY	DESCRIPTION
CS	ConfigSet	CS <key> <value>	–	Persist key to mc.ini; silent ok. axis2_use / WDT_use need RB to take HW effect.
CR	ConfigReset	CR	–	Reset all config to compiled defaults and save mc.ini.
CG	ConfigGet	CG [<key>]	CG: <key>= <value>	One key, or bare dumps all keys (multi-line).
RB	Reboot	RB	–	Soft MCU reset (no power cycle); EN off first. After CS axis2_use.

W — Wait

SHORT	LONG	CALL	REPLY	DESCRIPTION
W	Wait	W [<sec>]	—	Delay then continue ; chain; bare→1 s; never !E:timeout.
WM	WaitMoving	WM [<timeout_s>]	—	Pause chain until move ends; optional timeout → !E:timeout, cancel rest of chain.
WH	WaitHoming	WH [<timeout_s>]	—	Pause until homing ends; timeout same as WM.

V — Version

SHORT	LONG	CALL	REPLY	DESCRIPTION
VA	VersionAbout	VA	VA:...	About string (name, version, author).
VF	VersionFW	VF	VF: <version>	Firmware version.
VP	VersionProtocol	VP	VP:<n>	Protocol version.

Special

SHORT	LONG	CALL	REPLY	DESCRIPTION
H/HT	Halt	H HT	—	Immediate STEP abort; enable off; cancel waits and remaining ; chain.
P	Pins	VG P	VG:PIN_*=n (multi-line)	Machine-readable pin map (alias VersionGPIO). Axis-2 pins if axis2 on.
\$/HL	Help	\$ HL Help	(multi-line table)	ASCII table of all commands (≤80 columns).

Notes

Chain with ;. Realtime (no newline): ? status, ! soft stop, Ctrl-X soft reset.
 Path mode (PG): most move/session cmds → !E:busy; allowed:
 MS/H/RB/PD/PN/I*/G*/V*/IX/Help/CG.
 Soft limits / units: see config keys slider_min/max, steps_per_unit, unit_name.